Benjamin Warner

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Open to relocation (UK / EU) • Eligible for UK Youth Mobility visa — no employer sponsorship required

Summary

Fourth-year Computer Science student with two co-op terms at Ontario Power Generation, one of North America's largest clean-energy utilities, where I built AI tooling on Azure for enterprise engineering teams. I run my own multi-node infrastructure: Kubernetes with GitOps, Docker, Linux servers, private networking and local LLM inference. I want to build platforms that speed up the energy transition.

Experience

Computer Science Co-op — Software / AI May 2026 – Aug 2026
Ontario Power Generation *Oshawa, ON*

- Designed and built **OPGKit Studio**, an AI-powered backlog generator and **deterministic** development-time estimator for enterprise software teams.
- Generated structured tickets with **Azure AI Foundry** and stored their embeddings as **SQL vectors** for similarity search inside the existing enterprise database.
- Estimated effort by matching new tickets to past approved, estimated tickets by **cosine distance**, grounding AI output in the team's own history instead of model guesses.

Computer Science Co-op — Data & Digital Modernization Jan 2025 – Dec 2025
Ontario Power Generation *Darlington Nuclear & Oshawa, ON*

- **Maintenance Strategies (Darlington Nuclear):** built Power BI dashboards turning SCR and observation & coaching data into insights for plant maintenance teams.
- **Nuclear Modernization (Corporate HQ):** helped stand up the **Nucleus** platform and supported discovery for the Extrad application.
- Placed **2nd** in the Microsoft Copilot Hackathon (organized by OPG, held at Microsoft Canada HQ, Toronto).

Founder TODO: start year – Present
Axis Solutions — web & software agency for local small businesses *Durham Region, ON*

- Run the agency end to end (scoping, building, deploying, maintaining client sites), with an automated lead-outreach pipeline in self-hosted **n8n**.

Projects

Self-Hosted Platform Lab | *k3s, Argo CD, Docker, Tailscale, Cloudflare Tunnel, Nginx, Linux* 2024 – Present

- Run a multi-node lab (Mac Mini M1, Raspberry Pis, headless Ubuntu servers) joined in a private **Tailscale** mesh.
- Deploy apps **GitOps-style**: a **k3s** cluster syncs from GitHub through **Argo CD** and is published through **Cloudflare Tunnels**, with no open inbound ports.
- Operate containerized services behind **Nginx** with **Infisical** secrets, Postgres, Samba storage and n8n automation.
- Host **local LLM inference** (Ollama, mlx-lm) as a private endpoint, and run **Malcolm**, a fully local voice assistant (faster-whisper, Piper).

WeMake — realtime collaborative music studio in the browser | *Docker, GHCR, Cloudflare* 2026 – Present

- Built a realtime collaborative web app shipped as a **GHCR** container image, self-hosted on a Raspberry Pi and served publicly at wemake.ben-warner.dev via Cloudflare Tunnel.

MCP & AI Agent Tooling (Axis MCP, to-prod) | *Python, TypeScript, FastMCP, RAG, embeddings* 2026 – Present

- Building **axis-mcp-cli**, a management layer for installing and running Model Context Protocol (MCP) servers that connect AI agents to tools and data.
- Developing **to-prod**, which extends my OPG work into a product that sizes software projects from a team's historical tickets and real infrastructure, using retrieval (BM25 + embeddings) and constrained decoding to keep outputs grounded and deterministic.

Embedded Control Systems | *C/C++, ESP32, TMC2209, sensors* 2026 – Present

- **Automated greenhouse:** closed-loop ESP32 control of pump and fan from soil, light and humidity sensors.
- **Go-To telescope mount:** from-scratch ESP32 firmware implementing **LX200** and RA/Dec → alt/az → step pointing math.

Technical Skills

Cloud & Platform: Azure (AI Foundry), GCP, Cloudflare (Workers, Tunnels), Supabase, Railway, Coolify, Stripe metered billing
Infrastructure & DevOps: Kubernetes (k3s), Argo CD / GitOps, Docker & Compose, Nginx, Tailscale, Linux, systemd, Bash
AI & LLM: MCP / FastMCP, RAG, vector search, BM25, constrained decoding, agent loops, evals, Ollama, Claude & OpenAI APIs
Languages: Python, TypeScript, JavaScript, SQL, C/C++, Java, HTML/CSS
Frameworks & Data: Next.js, React, Angular, FastAPI, Node.js, Postgres, Power BI, Power Automate, Vitest

Education

Ontario Tech University Oshawa, ON
Bachelor of Science, Computer Science *Expected April 2027*

- Courses: Data Structures, Scientific Data Analytics, Computational Science, Environmental Science • CCC (2×)